Mahamat Nour Bachar Mahamat

PhD Candidate – Deep Learning for Medical Image Analysis & Healthcare

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Summary

My academic background in Data Science has fueled my research interest in computer vision and medical image analysis. I aim to pursue doctoral work that applies statistical learning and deep learning techniques to healthcare imaging challenges. My Master's research on image analysis, complemented by my initial medical studies (PACES), has cultivated a genuine interest in medical applications of imaging data. I'm particularly drawn to research with meaningful societal impact.

Professional Experience

Non-negative Matrix Factorization for Source Separation and Detection *May 2025 – Oct. 2025*

IMT Atlantique, Brest (***Research Internship***)

- Developed source separation methods for bioacoustic data based on NMF.
- Designed a detection pipeline for acoustic events combining NMF and temporal segmentation (`ruptures`).
- Explored sequential and «few-shot learning»-inspired approaches for detection.
- Work focused on component interpretability and computational efficiency.

Pharmaceutical products preparer (student job) – Alliance Healthcare *Dec. 2024 – Apr. 2025*

Order Picker (student job) – Kuehne+Nagel, Angers *May 2024 – Oct. 2024*

Versatile teammate(student work) – McDonald's Angers Saint-Serge *Sept. 2021 – Mar. 2024*

Education

Master's degree of Data Science, University of Angers *2023 – 2025*

Main courses (More details) :

- Image Processing with Deep Learning: classification, segmentation, object detection (mark: 20/20)
- Nonlinear Optimization & Introduction to signal processing
- Statistical Learning: theories and applications of supervised and unsupervised models
- Object-Oriented Programming (Python) & Relational Databases (SQL)

Bachelor's degree of Mathematics, University of Angers *2020 – 2023*

- Theoretical courses in analysis, algebra, probability, and geometry.
- Bachelor's thesis: Necessary and sufficient condition on a real polynomial for real roots.

PluriPASS (formerly PACES), University of Angers *2019 – 2020*

Selective first year of health studies: introduction to medical sciences, biology, engineering sciences, and scientific methodology. (*Year successfully completed*)

Baccalaureate, Series D (Maths & Biology) – Lycée La Fidélité, Chad *2018*

National rank: 22^{nd}/79,136

Academic & personal projects

ECG signal denoising (Personal project) *Sept. 2025 - Dec. 2025*

- Simulated various physiological noises (Gaussian, electrode motion, muscle artifacts) on ECG signals.
- Design and training of a convolutional autoencoder and a diffusion model for ECG signal denoising and restoration.

Blood cell image segmentation *Feb. 2025*

- Develop and training of semantic segmentation models (U-Net trained from scratch vs. a pre-trained VGG16 encoder).
- Performance comparison using Dice and IoU metrics, with data augmentation to improve generalization.

Object detection using pre-trained & foundation models *Feb. 2025*

- Implemented object detection pipelines based on YOLO11.
- Fine-tuned the YOLO model to improve detection on the MinneApple dataset.
- Compared performance between the DINO and YOLO approaches.

Diffusion Generative Models (Master's Thesis, *Read the report*) *Nov. 2024 – Mar. 2025*

Key skills: generative models (GAN, VAE, diffusion), stochastic differential equations, time reversal

- Study of mathematical foundations (SDEs, time reversal) and implementation of practical applications.
- Comparative evaluation of diffusion, GAN, and VAE models on MNIST using the MML (Mean Maximum Logit) score.
- Critical analysis of the strengths and limitations for synthetic data generation.

Generative Models (GAN & VAE) *Nov. 2024 – Dec. 2024*

- Development and training of GAN and VAE models.
- Analysis of results and comparison of the performance of those two generative models.

Database and Web Application – Enron Dataset *Jan. 2024 – May 2024*

Key skills: Data structuring, visualization, PostgreSQL, Django, HTML

- Data extraction using regular expressions (regex) and population of a PostgreSQL database.

- Development of a Django web interface for interactive visualization.

Find more project details on my GitHub and personal website.

Skills

Technical

- **Programming:** Python, R, C++ (basics), SQL
- **Apprentissage profond**: Autoencoders, Transformers(notions), GAN, diffusion models
- **Vision & Images Processing:** data augmentation, denoising, segmentation (U-Net), object detection (YOLO)
- **Data Eng.:** Django/Flask, Web scraping, NoSQL(MongoDB)
- **MLOps (intro) :** Docker, Deployment, monitoring
- **Signal Processing:** sampling, Fourier transform, STFT, filtering
- **Machine Learning:** supervised & unsupervised learning, dimensionality reduction, clustering
- **Tools :** Git, Unix/Linux, LaTeX, PowerBI

Soft Skills

- Scientific rigor and autonomy
- Analytical and synthesis skills
- Curiosity and adaptability

Languages

- **French:** bilingual
- **English:** proficient (reading/writing), intermediate (speaking)
- **Arabic:** native / bilingual

References

- **Dr. David Rousseau**
  Professor – *University of Angers*
  Head of ImHorPhen team & Courses: image/signal processing
  Contact: david.rousseau@univ-angers.fr
- **Dr. Axel Marmoret**
  Researcher – *Lab-STICC (IMT Atlantique Brest)*
  Internship supervisor: Passive acoustic monitoring using NMF
  Contact: axel.marmoret@imt-atlantique.fr
- **Dr. Gilles Stupfler**
  Statistician researcher – *University of Angers*
  Responsible for M1 and teaching time series and statistics
  Contact: gilles.stupfler@univ-angers.fr
- **Dr. Sylvain Lamprier**
  Computer Science researcher – *University of Angers*
  Responsible for courses on generative models and NLP
  Contact: sylvain.lamprier@univ-angers.fr

Interests

Board games, historical documentaries, running.